Globins: an old family of heme proteins

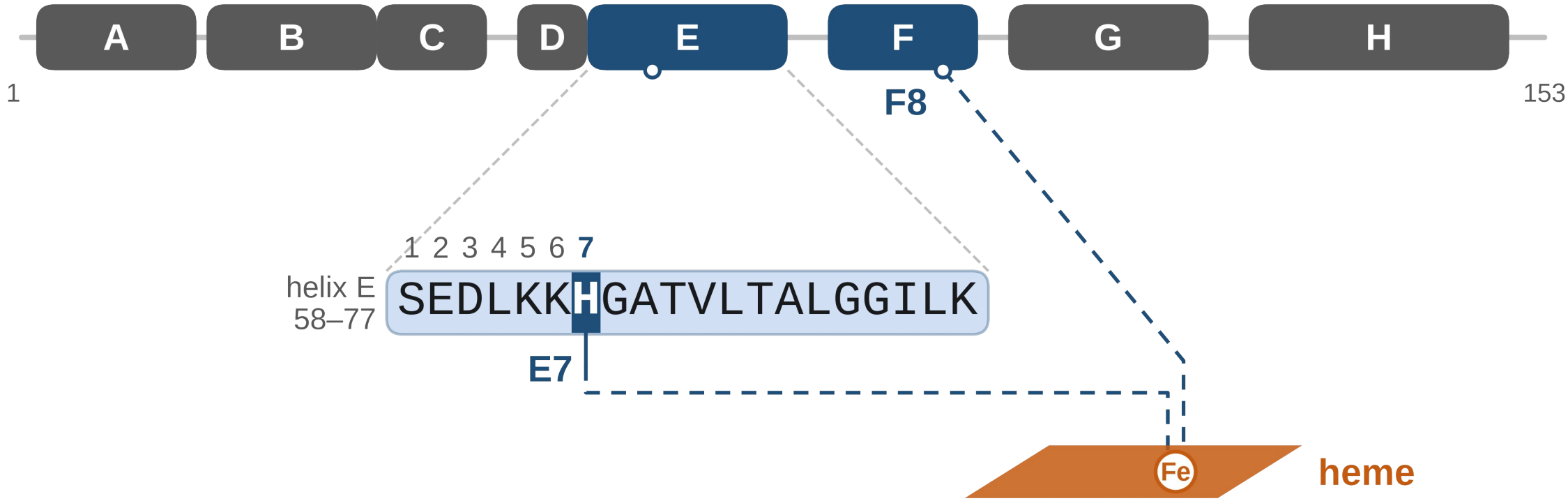
- Found in bacteria, archaea, fungi, plants and animals
- Functions: O2 transport and storage, NO detoxification, O2 sensing

	group	found in	function
M family myoglobin-like	**flavohemoglobins**	**bacteria,** fungi	**NO detoxification**
	plant globins	plants	
	hemoglobin α, β	animals	**O_2 transport**
	myoglobin	animals	**O_2 storage**
	neuroglobin	animals (brain and retina)	
	cytoglobin	animals	**NO dioxygenase**
S family sensors	**globin-coupled sensors**	**bacteria,** archaea	**O_2 sensing**
T family truncated fold	**truncated hemoglobins**	**bacteria,** archaea, plants, ciliates	

Vinogradov et al. 2013, BBA 1834:1789; Vinogradov et al. 2005, PNAS 102:11385; UniProt

The globin fold: eight helices A–H

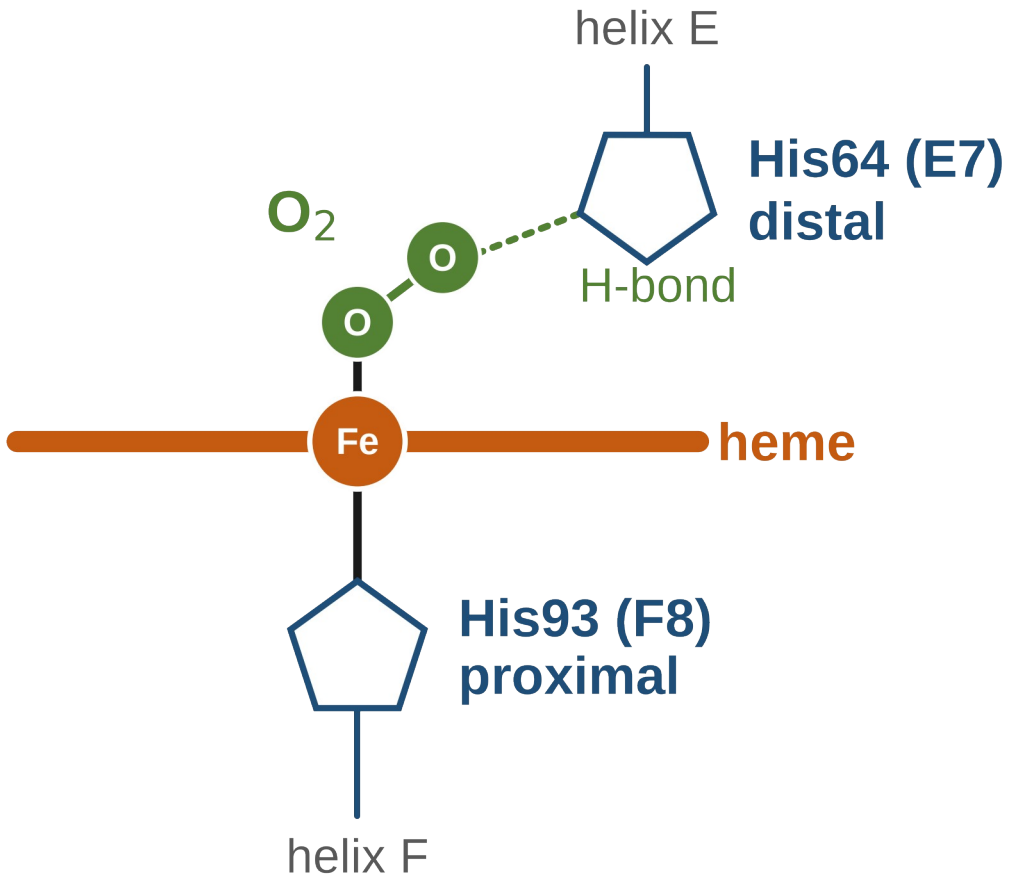
- The heme sits in a pocket between helices E and F
- Residues are named by helix and position: E7 = 7th residue of helix E



PDB 3RGK; UniProt P02144

Two histidines hold the heme iron

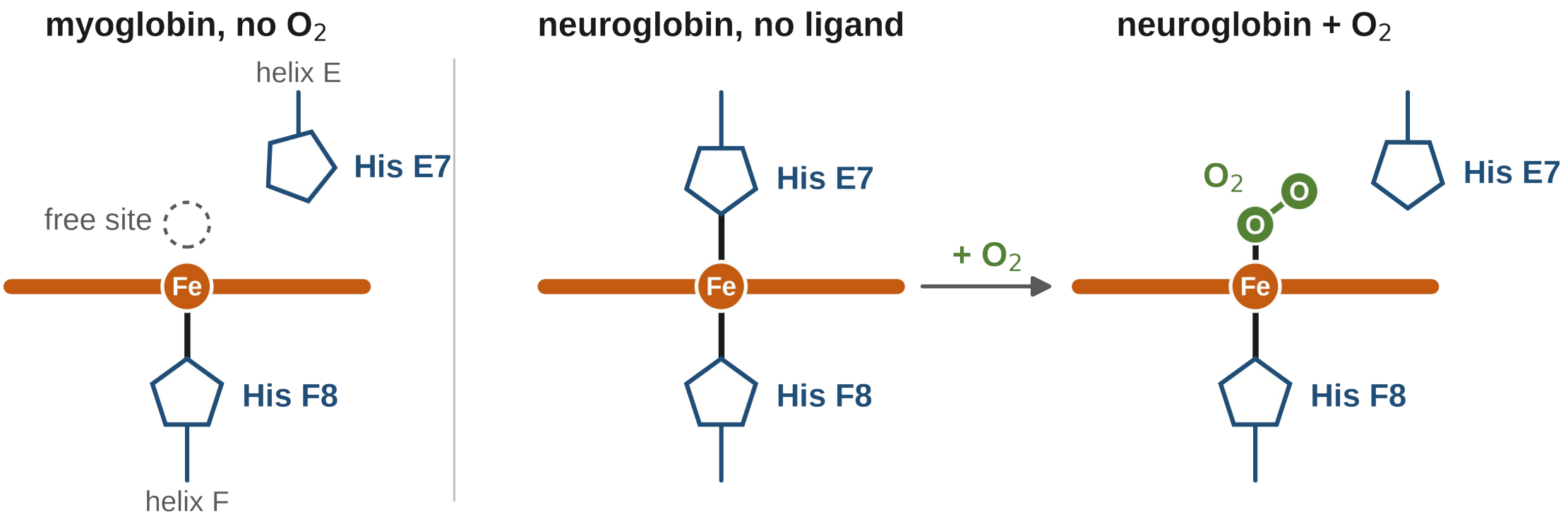
- **Proximal His (F8)** binds the iron from below
- **Distal His (E7)** sits above it, next to the bound O2
- E7 stabilises O2 by a hydrogen bond



Yang & Phillips 1996, J Mol Biol 256:762; Olson 2020, Antioxid Redox Signal 32:228

Neuroglobin: the distal His binds the iron

- Myoglobin without O2: iron has a free site (**pentacoordinate**)
- Neuroglobin without ligand: distal His binds the iron (**hexacoordinate**)
- A ligand must first push the distal His away



Pesce et al. 2003, Structure 11:1087 (PDB 1OJ6)

Finding E7 and F8 in an alignment



UniProt P69905, P68871, P02144, Q9NPG2, P01942, P02088, P04247, Q9ER97; PDB 3RGK; Clustal Omega